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SEMICONDUCTOR DEVICE HAVING INFORMATION STORAGE STRUCTURE

Abstract

A semiconductor device includes a lower structure having a conductive region, and an information storage structure on the lower structure. The information storage structure includes a lower electrode electrically connected to the conductive region, a support layer on a side surface of the lower electrode, a dielectric layer on the lower electrode and the support layer, an interface layer between the lower electrode and the dielectric layer, passivation layers between the support layer and the dielectric layer, and an upper electrode on the dielectric layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims benefit of priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0031201 filed on Mar. 5, 2024 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] The present inventive concept relates to a semiconductor device having an information storage structure.

[0003] As demand for implementation of high performance, high speed, and/or multifunctionalization of semiconductor devices increases, a degree of integration of semiconductor devices has been increasing. In manufacturing semiconductor devices having a fine pattern corresponding to the trend for a high degree of integration of semiconductor devices, it is necessary to implement patterns having a fine width or a fine isolation distance.

SUMMARY

[0004] An aspect of the present inventive concept provides a semiconductor device including an information storage structure.

[0005] According to an aspect of the present inventive concept, there is provided a semiconductor device including a lower structure having a conductive region, and an information storage structure on the lower structure. The information storage structure may include a lower electrode electrically connected to the conductive region, a support layer on a side surface of the lower electrode, a dielectric layer covering the lower electrode and the support layer, an interface layer between the lower electrode and the dielectric layer, passivation layers between the support layer and the dielectric layer, and an upper electrode on the dielectric layer.

[0006] According to another aspect of the present inventive concept, there is provided a semiconductor device including a lower structure having conductive regions, and an information storage structure on the lower structure. The information storage structure may include lower electrodes electrically connected to the conductive regions, a support layer between the lower electrodes, the support layer having a support hole (i.e., opening), a dielectric layer covering the lower electrodes and the support layer, interface layers between the lower electrodes and the dielectric layer, a passivation layer between the support layer and the dielectric layer, and an upper electrode on the dielectric layer. In plan view, the passivation layer may extend along a side surface of the support layer exposed by the support, between the lower electrodes. The interface layers may extend along side surfaces of the lower electrodes exposed by the support hole.

[0007] According to another aspect of the present inventive concept, there is provided a semiconductor device including a lower structure having a conductive region, and an information storage structure on the lower structure. The information storage structure may include a lower electrode electrically connected to the conductive region, a first support layer and a second support layer on a side surface of the lower electrode, the second support layer below the first support layer, a dielectric layer covering the lower electrode, the first support layer, and the second support layer, an interface layer between the lower electrode and the dielectric layer, a first passivation layer between the first support layer and the dielectric layer, a second passivation layer between the second support layer and the dielectric layer, and an upper electrode on the dielectric layer. The interface layer may have a first portion in contact with an upper surface of the lower electrode and the first passivation layer, and a second portion between the first support layer and the second support layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The above and other aspects, features, and advantages of the present inventive concept will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, wherein like reference numerals (when used) indicate corresponding elements throughout the several views, and in which:

[0009] FIG. **1** is a schematic plan view of a semiconductor device according to an example embodiment;

[0010] FIG. **2** is a schematic vertical cross-sectional view taken along lines II-I' and II-II' of the semiconductor device illustrated in FIG. **1**;

[0011] FIG. **3** is an enlarged view of a portion of the semiconductor device illustrated in FIG. **1**;

[0012] FIGS. **4** and **5** are partially enlarged views of the semiconductor device illustrated in FIG. **2**;

[0013] FIGS. **6** to **8** are schematic vertical cross-sectional views of semiconductor devices according to example embodiments;

[0014] FIG. **9** is a flowchart of a method of forming a semiconductor device according to an example embodiment;

[0015] FIGS. **10** to **16** are schematic vertical cross-sectional views of sequential intermediate processes of a method of manufacturing a semiconductor device according to an example embodiment;

[0016] FIG. **17** is a flowchart of a method of forming an interface layer according to an example embodiment;

[0017] FIGS. **18** to **20** are schematic vertical cross-sectional views of sequential intermediate processes of a method of forming an interface layer according to an example embodiment;

[0018] FIG. **21** is a flowchart of a method of forming an interface layer according to an example embodiment;

[0019] FIG. **22** illustrates concentrations of an interface layer according to comparative examples and examples of the present inventive concept;

[0020] FIG. **23** illustrates defect rates according to comparative examples and examples of the present inventive concept;

[0021] FIG. **24** is a schematic plan view of an integrated circuit device according to example embodiments; and

[0022] FIG. **25** is a schematic vertical cross-sectional view taken along lines X1-X1' and Y1-Y1' of the integrated circuit device illustrated in FIG. **24**.

DETAILED DESCRIPTION

[0023] Hereinafter, preferred example embodiments of the present inventive concept will be described with reference to the attached drawings.

[0024] FIG. **1** is a schematic plan view of a semiconductor device according to an example embodiment. FIG. **2** is a schematic vertical cross-sectional view taken along lines I-I' and II-II' of the semiconductor device illustrated in FIG. **1**.

[0025] Referring to FIGS. **1** and **2**, a semiconductor device **100** may include a substrate **101** having active regions ACT, a device isolation layer **110** defining the active regions ACT in the substrate **101**, a word line structure WLS buried in the substrate **101**, the word line structure WLS including a word line WL, a bit line structure BLS extending to intersect the word line structure WLS on the substrate **101**, the bit line structure BLS including a bit line BL, and an information storage structure CAP on the bit line structure BLS. The information storage structure CAP may store information, and may be, for example, a capacitor structure of a dynamic random access memory (DRAM). The semiconductor device **100** may further include a lower conductive pattern **150** on the active region ACT, an upper conductive pattern **160** on the lower conductive pattern **150**, and

an insulating pattern **165** passing through the upper conductive pattern **160**.

[0026] The semiconductor device **100** may include, for example, a cell array of the DRAM. For example, the bit line BL may be connected to a first impurity region **105a** of the active region ACT, and a second impurity region **105b** of the active region ACT may be electrically connected to the information storage structure CAP on the upper conductive pattern **160** through the lower and upper conductive patterns **150** and **160**.

[0027] The information storage structure CAP may be a capacitor capable of storing

[0028] information in a memory such as the DRAM. The information storage structure CAP may be electrically connected to the conductive regions **150** and **160**, on a lower structure including the lower and upper conductive patterns **150** and **160**. Here, the lower structure may include a substrate **101**, a word line structure WLS, and a bit line structure BLS.

[0029] The information storage structure CAP may include lower electrodes **170**, a dielectric layer **180** on the lower electrodes **170**, interface layers **175** between the lower electrodes **170** and the dielectric layer **180**, and an upper electrode **190** on the dielectric layer **180**. The information storage structure CAP may further include support layers SP and passivation layers PL.

[0030] The semiconductor device **100** may include a cell array region in which a cell array is disposed, and a peripheral circuit region in which peripheral circuits for driving memory cells, disposed in the cell array, are disposed. The peripheral circuit region may be disposed around the cell array region.

[0031] The substrate **101** may include a semiconductor material, such as a group IV semiconductor, a group III-V compound semiconductor, or a group II-VI compound semiconductor. For example, the group IV semiconductor may include silicon, germanium, or silicon-germanium. The substrate **101** may further include impurities. The substrate **101** may be a silicon substrate, a silicon-on-insulator (SOI) substrate, a germanium substrate, a germanium-on-insulator (GOI) substrate, a silicon-germanium substrate, or a substrate including an epitaxial layer.

[0032] Active regions ACT may be defined by the device isolation layer **110** in the substrate **101**. The active region ACT may have a bar shape, and may be disposed to have an island shape extending in one direction in the substrate **101**. The one direction may be inclined with respect to a direction in which the word lines WL and the bit lines BL extend. The active regions ACT may be disposed to be parallel to each other, and an end of one active region ACT may be disposed to be adjacent to the center of another active region ACT adjacent thereto.

[0033] The active region ACT may have first and second impurity regions **105a** and **105b** having a predetermined depth from an upper surface of the substrate **101**. The first and second impurity regions **105a** and **105b** may be spaced apart from each other. The first and second impurity regions **105a** and **105b** may serve as source/drain regions of a transistor formed by the word line WL. The source region and the drain region, formed by the first and second impurity regions **105a** and **105b** caused by doping or ion implantation of substantially the same impurities, may be interchangeably referred to depending on a circuit configuration of a finally formed transistor. The impurities may include impurities having a conductivity type opposite to that of the substrate **101**. In example embodiments, the depths of the first and second impurity regions **105a** and **105b** may be different from each other in the source region and the drain region.

[0034] The device isolation layer **110** may be formed using a shallow trench isolation (STI) process. The device isolation layer **110** may electrically isolate the active regions ACT from each other while surrounding the active regions ACT. The device isolation layer **110** may be made of an insulating material, for example, silicon oxide, silicon nitride, or a combination thereof. The device isolation layer **110** may have a plurality of regions having different lower end depths depending on a width of a trench in which the substrate **101** is etched.

[0035] The word line structures WLS may be disposed in gate trenches **115** extending in the substrate **101**. Each of the word line structures WLS may include a gate dielectric layer **120**, a word line WL, and a gate capping layer **125**. As used herein, a “gate **120** and WL” may be referred to as

a structure including the gate dielectric layer **120** and the word line WL, the word line WL may be referred to as a “gate electrode,” and the word line structure WLS may be referred to as a “gate structure.”

[0036] The word line WL may extend in a first direction X across the active region ACT. For example, a pair of word lines WL, adjacent to each other, may be disposed to cross one active region ACT. The word line WL may form a gate of a buried channel array transistor (BCAT), but the present inventive concept is not limited thereto. In example embodiments, the word lines WL may be disposed on an upper portion of the substrate **101**. The word line WL may be disposed on a lower portion of a gate trench **115** to have a predetermined thickness. An upper surface of the word line WL may be positioned at a level lower than that of the upper surface of the substrate **101**. As used herein, the high and the low of the term “level” may be defined based on a substantially flat (i.e., planar) upper surface of the substrate **101**.

[0037] The word line WL may include a conductive material, for example, at least one of polycrystalline silicon (Si), titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (TaN), tungsten (W), tungsten nitride (WN), and aluminum (Al). For example, the word line WL may include a lower pattern and an upper pattern, formed of different materials. The lower pattern may include at least one of tungsten (W), titanium (Ti), tantalum (Ta), tungsten nitride (WN), titanium nitride (TiN), and tantalum nitride (TaN). The upper pattern may be a semiconductor pattern including polysilicon doped with a P-type or N-type impurity.

[0038] The gate dielectric layer **120** may be disposed on a bottom surface and internal side surfaces (i.e., sidewalls) of the gate trench **115**. The gate dielectric layer **120** may conformally cover an internal wall of the gate trench **115**. The term “conformally” (or “conformal,” or like terms), as may be used herein in the context of a material layer or coating, is intended to refer broadly to a material layer or coating having a substantially uniform cross-sectional thickness relative to the contour of a surface to which the material layer is applied. The term “cover” (or “covers” or “covering,” or like terms), as may be used herein, is intended to broadly refer to a material, layer or structure being on or over another material, layer or structure, but does not require the material, layer or structure to entirely cover the other material, layer or structure. The gate dielectric layer **120** may include at least one of silicon oxide, silicon nitride, and silicon oxynitride. The gate dielectric layer **120** may be, for example, a silicon oxide film or an insulating film having a high dielectric constant. In example embodiments, the gate dielectric layer **120** may be a layer formed by oxidizing the active region ACT or a layer formed by deposition.

[0039] The gate capping layer **125** may be disposed to fill the gate trench **115**, on an upper portion of the word line WL. The term “fill” (or “filling,” “filled,” or like terms), as may be used herein, is intended to refer broadly to either completely filling a defined space (e.g., the gate trench **115**) or partially filling the defined space; that is, the defined space need not be entirely filled but may, for example, be partially filled or have voids or other spaces throughout. An upper surface of the gate capping layer **125** may be positioned at a level substantially the same as that of the upper surface of the substrate **101**; that is, the upper surface of the gate capping layer **125** may be coplanar with the upper surface of the substrate **101**. The gate capping layer **125** may be formed of an insulating material, for example, silicon nitride.

[0040] The bit line structure BLS may extend in a direction, perpendicular to the word line WL, for example, a second direction Y. The bit line structure BLS may include a bit line BL, and a bit line capping pattern BC on the bit line BL.

[0041] The bit line BL may include a first conductive pattern **141**, a second conductive pattern **142**, and a third conductive pattern **143**, sequentially stacked in the Z-direction (i.e., vertically). The bit line capping pattern BC may be disposed on the third conductive pattern **143**. A buffer insulating layer **128** may be disposed between the first conductive pattern **141** and the substrate **101**, and a portion of the first conductive pattern **141** (hereinafter, a bit line contact pattern DC) may be in contact with the first impurity region **105a** of the active region ACT. The term “contact” (or

“contacting,” or like terms, such as “connect” or “connecting”), as may be used herein, is intended to refer to a physical and/or electrical connection between two or more elements, and may include other intervening elements. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. The bit line BL may be electrically connected to the first impurity region **105a** through the bit line contact pattern DC. A lower surface of the bit line contact pattern DC may be positioned at a level lower than that of the upper surface of the substrate **101**, and may be positioned at a level higher than that of the upper surface of the word line WL. In an example embodiment, the bit line contact pattern DC may be formed in the substrate **101** to be locally disposed in a bit line contact hole, exposing the first impurity region **105a**.

[0042] The first conductive pattern **141** may include a semiconductor material such as polycrystalline silicon. The first conductive pattern **141** may be in direct contact with the first impurity region **105a**. The second conductive pattern **142** may include a metal-semiconductor compound. The metal-semiconductor compound may be, for example, a layer obtained by silicidizing a portion of the first conductive pattern **141**. For example, the metal-semiconductor compound may include cobalt silicide (CoSi), titanium silicide (TiSi), nickel silicide (NiSi), tungsten silicide (WSi), or other metal silicides. The third conductive pattern **143** may include a metal material such as titanium (Ti), tantalum (Ta), tungsten (W), and aluminum (Al). In some example embodiments, the number of conductive patterns included in the bit line BL, a type of material, and/or a stacking order may be changed in various manners.

[0043] The bit line capping pattern BC may include a first capping pattern **146**, a second capping pattern **147**, and a third capping pattern **148**, sequentially stacked on the third conductive pattern **143** in the Z-direction. Each of the first to third capping patterns **146**, **147**, and **148** may include an insulating material, for example, a silicon nitride film. The first to third capping patterns **146**, **147**, and **148** may be formed of different materials. Even when the first to third capping patterns **146**, **147**, and **148** include the same material, the first to third capping patterns **146**, **147**, and **148** may be distinguished from each other due to a difference in physical properties. A cross-sectional thickness of the second capping pattern **147** may be less than a cross-sectional thickness of the first capping pattern **146** and a cross-sectional thickness of the third capping pattern **148**, respectively. In some example embodiments, the number of capping patterns and/or a type of material, included in the bit line capping pattern BC, may be changed in various manners.

[0044] Spacer structures SS may be disposed on opposite sidewalls of each of the bit line structures BLS to extend in a direction, for example, a Y-direction. The spacer structures SS may be disposed between the bit line structure BLS and the lower conductive pattern **150**. The spacer structures SS may extend along sidewalls of the bit line BL and sidewalls of the bit line capping pattern BC. Fence insulating patterns **154** may be formed between the spacer structures SS. The fence insulating patterns **154** may include silicon nitride or silicon oxynitride. A pair of spacer structures SS, disposed on opposite sides of one bit line structure BLS, may have an asymmetric shape with respect to the bit line structure BLS. Each of the spacer structures SS may include a plurality of spacer layers, and may further include an air spacer in some example embodiments.

[0045] The lower conductive pattern **150** may be connected to one region of the active region ACT, for example, the second impurity region **105b**. The lower conductive pattern **150** may be disposed between the bit lines BL and between the word lines WL. The lower conductive pattern **150** may pass through the buffer insulating layer **128** to be connected to the second impurity region **105b** of the active region ACT. The lower conductive pattern **150** may be in direct contact with the second impurity region **105b**. A lower surface of the lower conductive pattern **150** may be positioned at a level lower than that of the upper surface of the substrate **101**, and may be positioned at a level higher than that of the lower surface of the bit line contact pattern DC. The lower conductive pattern **150** may be insulated from the bit line contact pattern DC by the spacer structure SS. The lower conductive pattern **150** may be formed of a conductive material. For example, the conductive material may include at least one of polycrystalline silicon (Si), titanium (Ti), titanium nitride

(TiN), tantalum (Ta), tantalum nitride (TaN), tungsten (W), tungsten nitride (WN), and/or aluminum (Al), although embodiments are not limited thereto. In example embodiments, the lower conductive pattern **150** may include a plurality of layers.

[0046] A metal-semiconductor compound layer **155** may be disposed between the lower conductive pattern **150** and the upper conductive pattern **160**. The metal-semiconductor compound layer **155** may be, for example, a layer obtained by silicidizing a portion of the lower conductive pattern **150**, when the lower conductive pattern **150** includes a semiconductor material. The metal-semiconductor compound layer **155** may include, for example, cobalt silicide (CoSi), titanium silicide (TiSi), nickel silicide (NiSi), tungsten silicide (WSi), or other metal silicides. In some example embodiments, the metal-semiconductor compound layer **155** may be omitted.

[0047] The upper conductive pattern **160** may be disposed on the lower conductive pattern **150**. The upper conductive pattern **160** may extend between the spacer structures SS to cover an upper surface of the metal-semiconductor compound layer **155**. The upper conductive pattern **160** may include a barrier layer **162** and a conductive layer **164**. The barrier layer **162** may cover (e.g., conformally cover) a lower surface and side surfaces of the conductive layer **164**. The barrier layer **162** may include a metal nitride, for example, at least one of titanium nitride (TiN), tantalum nitride (TaN), and/or tungsten nitride (WN), although embodiments are not limited thereto. The conductive layer **164** may include a conductive material, for example, at least one of polycrystalline silicon (Si), titanium (Ti), tantalum (Ta), tungsten (W), ruthenium (Ru), copper (Cu), molybdenum (Mo), platinum (Pt), nickel (Ni), cobalt (Co), aluminum (Al), titanium nitride (TiN), tantalum nitride (TaN), and/or tungsten nitride (WN).

[0048] Insulating patterns **165** may be disposed between the upper conductive patterns **160**. The insulating patterns **165** may electrically insulate the upper conductive patterns **160** from each other. The insulating patterns **165** may include an insulating material, for example, at least one of silicon oxide, silicon nitride, and/or silicon oxynitride.

[0049] The etch stop layer **168** may cover the insulating patterns **165**, between the first electrode structures **170**. The etch stop layer **168** may be in contact with lower regions of side surfaces of the first electrode structures **170**. The etch stop layer **168** may be disposed below the support layers SP. The etch stop layer **168** may include, for example, at least one of silicon nitride (SiN), silicon oxynitride (SiON), silicon carbonitride (SiCN), and/or silicon boron nitride (SiBN).

[0050] FIG. **3** is an enlarged view of a portion of the semiconductor device illustrated in FIG. **1**. FIGS. **4** and **5** are partially enlarged views of the semiconductor device illustrated in FIG. **2**.

[0051] Referring further to FIGS. **3** to **5**, the lower electrodes **170** may be disposed on the upper conductive patterns **160**. The lower electrodes **170** may have a pillar shape. The lower electrodes **170** may be spaced apart from each other in an X-direction and a Y-direction. In an example embodiment, the lower electrodes **170** may be arranged to have a honeycomb structure. For example, as illustrated in FIG. **1**, the lower electrodes **170** may be disposed at vertices of a hexagonal pattern and the center of the hexagonal pattern, respectively.

[0052] As illustrated in FIG. **4**, in an example embodiment, the lower electrode **170** may include a first layer **171** and a second layer **172**. The second layer **172** may cover or surround the first layer **171**, and may include metal oxide and metal oxynitride. For example, the first layer **171** may include at least one of vanadium nitride (VN), titanium nitride (TiN), titanium silicide nitride (TiSiN), niobium nitride (NbN), molybdenum nitride (MoN), tantalum nitride (TaN), ruthenium (Ru), ruthenium oxide (RuO₂), platinum (Pt), iridium (Ir), strontium ruthenate (SrRuO₃), tungsten (W), and tungsten nitride (WN), although embodiments are not limited thereto. The second layer **172** may include at least one of niobium oxide (NbOx), niobium oxynitride (NbON), tantalum oxide (TaOx), and tantalum oxynitride (TaON).

[0053] The support layers SP may include a first support layer SP1, a second support layer SP2 disposed below the first support layer SP1, and a third support layer SP3 disposed below the second support layer SP2. The support layers SP1, SP2, and SP3 may be spaced apart from the

substrate **101** in a Z-direction, perpendicular to an upper surface of the substrate **101**. The support layers **SP1**, **SP2**, and **SP3** may be in contact with the lower electrodes **170**, and may extend in a direction (horizontal direction), parallel to the upper surface of the substrate **101**.

[0054] The first support layer **SP1** may have a cross-sectional thickness greater than those of the second support layer **SP2** and the third support layer **SP3**, but the present inventive concept is not limited thereto. The support layers **SP1**, **SP2**, and **SP3** may be layers supporting the lower electrodes **170** having a high aspect ratio. The support layers **SP1**, **SP2**, and **SP3** may respectively include, for example, at least one of silicon nitride (**SiN**), silicon oxynitride (**SiON**), silicon carbonitride (**SiCN**), and/or silicon boron nitride (**SiBN**). In an example embodiment, the support layers **SP1**, **SP2**, and **SP3** may include silicon carbonitride (**SiCN**). The number and cross-sectional thicknesses of the support layers **SP1**, **SP2**, and **SP3**, and/or an arrangement relationship between the support layers **SP1**, **SP2**, and **SP3** are not limited to those illustrated, and may be changed in various manners in some example embodiments.

[0055] An upper surface of the first support layer **SP1** may be coplanar with upper surfaces of the lower electrodes **170**, relative to the upper surface of the substrate **101** as a reference layer.

[0056] As illustrated in FIG. 1, the support layers **SP1**, **SP2**, and **SP3** may be disposed between the lower electrodes **170**, and may support the lower electrodes **170**. The support layers **SP1**, **SP2**, and **SP3** may respectively include support holes **SH**. The support holes **SH**, formed in the support layers **SP1**, **SP2**, and **SP3**, may overlap each other in a vertical direction (i.e., Z-direction), but the present inventive concept is not limited thereto. The term “overlap” (or “overlapping,” or like terms), as may be used herein, is intended to broadly refer to a first element that intersects with at least a portion of a second element in the vertical direction (i.e., Z-direction), but does not require that the first and second elements be completely aligned with one another in a horizontal plane (i.e., in the X-direction and/or the Y-direction). FIG. 1 illustrates that the support holes **SH** are disposed between four adjacent lower electrodes **170**, but the present inventive concept is not limited thereto. A shape and an arrangement method of the support hole **SH** may vary in some example embodiments.

[0057] The passivation layer **PL** may be disposed between the support layers **SP1**, **SP2**, and **SP3** and the dielectric layer **180**. The passivation layer **PL** may include a first passivation layer **PL1**, a second passivation layer **PL2**, and a third passivation layer **PL3**. The first passivation layer **PL1** may cover the first support layer **SP1**. For example, in cross-sectional view, the first passivation layer **PL1** may cover upper and lower surfaces of the first support layer **SP1**. In plan view, the first passivation layer **PL1** may cover a side surface of the first support layer **SP1** exposed by the support holes **SH**. The second passivation layer **PL2** and the third passivation layer **PL3** may have a structure the same as or similar to that of the first passivation layer **PL1**. For example, the second passivation layer **PL2** may cover upper and lower surfaces of the second support layer **SP2**, and may cover a side surface of the second support layer **SP2** exposed by the support holes **SH**. The first passivation layer **PL1**, the second passivation layer **PL2**, and the third passivation layer **PL3** may also be in contact with a side surface of the lower electrode **170**.

[0058] The passivation layer **PL** may include a material different from that of the support layer **SP**. For example, the passivation layer **PL** may include a halogen compound. The first passivation layer **PL1**, the second passivation layer **PL2**, and the third passivation layer **PL3** may respectively include at least one of fluorine (**F**), chlorine (**Cl**), bromine (**Br**), and/or iodine (**I**).

[0059] The semiconductor device **100** may further include a lower passivation layer **PLL**, disposed on the etch stop layer **168**. The lower passivation layer **PLL** may be disposed between the etch stop layer **168** and the dielectric layer **180**. The lower passivation layer **PLL** may be in contact with lower regions of side surfaces of the lower electrodes **170**. The lower passivation layer **PLL** may include a halogen compound.

[0060] The interface layer **175** may be disposed between the lower electrode **170** and the dielectric layer **180**. The interface layer **175** may include a first portion **175a**, a second portion **175b**, and a

third portion **175c**. The first portion **175a** of the interface layer **175** may cover an upper surface of the lower electrode **170**, and may extend in a horizontal direction. The first portion **175a** of the interface layer **175** may be in contact with a side surface of the first passivation layer **PL1**. It is illustrated that a thickness of the first portion **175a** of the interface layer **175** is greater than a thickness of the first passivation layer **PL1**, but the present inventive concept is not limited thereto. [0061] The second portion **175b** of the interface layer **175** may cover the side surface of the lower electrode **170**, and may extend in the vertical direction. The second portions **175b** may be respectively in contact with an upper surface or a lower surface of at least one of the first passivation layer **PL1**, the second passivation layer **PL2**, and the third passivation layer **PL3**. The second portions **175b** may overlap the support layers **SP1**, **SP2**, and **SP3** in the vertical direction (i.e., Z-direction).

[0062] The third portion **175c** of the interface layer **175** may cover the side surface of the lower electrode **170** exposed by the support holes **SH**, and may extend in the vertical direction. The second portion **175b** and the third portion **175c** of the interface layer **175** may be in contact with an upper surface of the lower passivation layer **PLL**. Lower surfaces of the second portion **175b** and the third portion **175c** may be respectively positioned at a level higher than that of a lower surface of the lower electrode **170**.

[0063] The interface layer **175** may include a conductive material. In an example embodiment, the interface layer **175** may include metal oxide or metal oxynitride, and may include, as a metal element, at least one of scandium (Sc), yttrium (Y), titanium (Ti), vanadium (V), niobium (Nb), tantalum (Ta), chromium (Cr), molybdenum (Mo), tungsten (W), iron (Fe), ruthenium (Ru), osmium (Os), cobalt (Co), rhodium (Rh), iridium (Ir)), boron (B), tin (Sn), platinum (Pt), and/or lanthanum (La). In an example embodiment, the interface layer **175** may include at least one of niobium oxide (NbOx) and tantalum oxide (TaOx). The interface layer **175** may improve capacitance of the information storage structure **CAP**.

[0064] The dielectric layer **180** may cover the lower electrodes **170** and the support layers **SP1**, **SP2**, and **SP3**. The dielectric layer **180** may be disposed between the lower electrodes **170** and the upper electrode **190**. The dielectric layer **180** may be in contact with the passivation layer **PL** and the interface layer **175**.

[0065] The dielectric layer **180** may include a high dielectric constant (high-K) material, silicon oxide, silicon nitride, or a combination thereof. However, in some example embodiments, the dielectric layer **180** may include oxide, nitride, silicide, oxynitride, or silicon oxynitride including at least one or combinations of titanium (Ti), tantalum (Ta), hafnium (Hf), aluminum (Al), zirconium (Zr), and/or lanthanum (La), doped with fluorine (F). The dielectric layer **180** may further include one or more elements, among aluminum (Al), yttrium (Y), vanadium (V), and silicon (Si), in the form of a solid solution or an oxide film. In an example embodiment, the semiconductor device **100** may further include a capping layer disposed on an upper surface or a lower surface of the dielectric layer **180**, or may further include an insertion film disposed in the dielectric layer **180**. Main elements of the capping layer and the insertion layer may include at least one of silicon (Si), boron (B), lithium (Li), scandium (Sc), strontium (Sr), aluminum (Al), yttrium (Y), niobium (Nb), tantalum (Ta), titanium (Ti), molybdenum (Mo), and lanthanum (La).

[0066] The upper electrode **190** may cover a plurality of lower electrodes **170**, the support layers **SP1**, **SP2**, and **SP3**, and the dielectric layer **180**. The upper electrode **190** may at least partially fill a space between the plurality of lower electrodes **170**, and a space between the support layers **SP1**, **SP2**, and **SP3**. The upper electrode **190** may be in direct contact with the dielectric layer **180**. In an example embodiment, the upper electrode **190** may include at least one of TiN, NbN, WN, VN, MON, TaN, TiSiN, TiCN, a silicon material, and a silicon-germanium material.

[0067] Referring further to FIG. 3, in plan view, a side surface of the support layer **SP** and side surfaces of the lower electrodes **170** may be exposed by the support hole **SH**. The passivation layers **PL** may extend along an internal wall of the support hole **SH**, between the lower electrodes

170. For example, the passivation layers PL may extend along the side surface of the support layer SP exposed by the support hole SH, and may have an arc shape. The interface layers **175** may extend along the internal wall of the support hole SH, between the passivation layers PL. For example, the interface layers **175** may extend along the side surfaces of the lower electrodes **170** exposed by the support hole SH, and may have an arc shape. The lower electrodes **170** may respectively have a first portion **170a** not exposed by the support hole SH, and a second portion **170b** and a third portion **170c** exposed by the support hole SH. The first portion **170a** may be in contact with the support layer SP, the second portion **170b** may be in contact with the passivation layer PL, and the third portion **170c** may be in contact with the interface layer **175**. The second portion **170b** may be disposed between the first portion **170a** and the third portion **170c**. In plan view, a portion of the passivation layers PL may be in contact with the interface layers **175**. For example, a portion of the passivation layers PL may be covered by the interface layers **175** and spaced apart from the dielectric layer **180**.

[0068] FIGS. **6** to **8** are schematic vertical cross-sectional views of semiconductor devices according to example embodiments.

[0069] Referring to FIG. **6**, a semiconductor device **100a** may include a passivation layer PL (PL**1**) on a support layer SP (SP**1**), and an interface layer **175** (**175a**) on a lower electrode **170**. In an example embodiment, the passivation layer PL may not have a constant cross-sectional thickness. For example, the thickness of the passivation layer PL may decrease as a distance from the lower electrode **170** decreases. In an example embodiment, a portion of the interface layer **175** may overlap the passivation layer PL in a vertical direction (i.e., Z-direction). The interface layer **175** may not have a constant cross-sectional thickness. A portion of the interface layer **175** may extend onto the support layer SP and cover the passivation layer PL. The cross-sectional thickness of the interface layer **175** may decrease as a distance from the support layer SP decreases.

[0070] Referring to FIG. **7**, a semiconductor device **100b** may include a passivation layer PL on a support layer SP, and an interface layer **175** on a lower electrode **170**. In an example embodiment, the passivation layer PL may extend further onto the lower electrode **170**. For example, a portion of the passivation layer PL may be in contact with the lower electrode **170**, and may be disposed between the interface layer **175** and the lower electrode **170**.

[0071] Referring to FIG. **8**, a semiconductor device **100c** may include a passivation layer PL on a support layer SP, and an interface layer **175** on a lower electrode **170**. In an example embodiment, the interface layer **175** may extend further onto the passivation layer PL. For example, the passivation layer PL may be covered by the interface layer **175**, and the interface layer **175** may extend in a horizontal direction, between adjacent lower electrodes **170**. In an example embodiment, a cross-sectional thickness of a portion of the interface layer **175**, covering the passivation layer PL, may be less than a cross-sectional thickness of a portion of the interface layer **175**, covering the lower electrodes **170**. The interface layer **175** may be interposed between the passivation layer PL and a dielectric layer **180**, and the passivation layer PL may be spaced apart from the dielectric layer **180**.

[0072] FIG. **9** is a flowchart of a method of forming a semiconductor device according to an example embodiment.

[0073] Referring to FIG. **9**, a method of manufacturing a semiconductor device according to an example embodiment may include forming a lower structure and a mold structure on a substrate (S**10**), forming a plurality of holes passing through the mold structure (S**20**), forming lower electrodes in a plurality of holes (S**30**), removing a portion of the mold structure to form a support layer (S**40**), forming a passivation layer on the support layer and forming an interface layer on the lower electrodes (S**50**), and forming a dielectric layer and an upper electrode (S**60**).

[0074] FIGS. **10** to **16** are schematic vertical cross-sectional views of sequential intermediate processes of a method of manufacturing a semiconductor device according to an example embodiment.

[0075] Referring to FIGS. **9** and **10**, a lower structure and a mold structure ST may be formed on a substrate (**S10**). First, a device isolation layer **110** may be formed on a substrate **101** to define an active region ACT. A device isolation trench may be formed in the substrate **101**, and the device isolation layer **110** may fill the device isolation trench. In plan view, the active region ACT may have a shape of an elongated bar extending in a direction, diagonal to a direction in which a word line WL extends. An ion implantation process may be performed using the device isolation layer **110** as an ion implantation mask to form impurity regions on an upper portion of the active region ACT. A gate trench **115** may be formed by patterning the active region ACT and the device isolation layer **110**. A pair of gate trenches **115** may cross the active region ACT, but the present inventive concept is not limited thereto. The impurity regions may also be isolated from each other by the gate trench **115** to form a first impurity region **105a** and a second impurity region **105b**.

[0076] A gate dielectric layer **120** may be formed on an internal surface of the gate trench **115** to have a substantially conformal (i.e., uniform) thickness. Subsequently, the word line WL may be formed to fill at least a portion of the gate trench **115**. An upper surface of the word line WL may be recessed to be lower than an upper surface of the active region ACT. An insulating layer may be stacked on the substrate **101** to fill the gate trench **115** and then etched to form a gate capping layer **125** on the word line WL.

[0077] An insulating layer and a conductive layer may be sequentially formed and patterned on a front surface of the substrate **101** to form a buffer insulating layer **128** and a first conductive pattern **141**, sequentially stacked. The buffer insulating layer **128** may be formed of at least one of silicon oxide, silicon nitride, and silicon oxynitride. A plurality of buffer insulating layers **128** may be spaced apart from each other. The first conductive pattern **141** may have a shape corresponding to a planar shape of the buffer insulating layer **128**. The buffer insulating layer **128** may be formed to simultaneously cover ends of two adjacent active regions ACT, that is, adjacent second impurity regions **105b**. A bit line contact hole may be formed by etching upper portions of the device isolation layer **110**, the substrate **101**, and the gate capping layer **125** using the buffer insulating layer **128** and the first conductive pattern **141** as an etching mask. The bit line contact hole may expose the first impurity region **105a**.

[0078] A bit line contact pattern DC, filling the bit line contact hole, may be formed. Forming the bit line contact pattern DC may include forming a conductive layer filling the bit line contact hole, and performing a planarization process. For example, the bit line contact pattern DC may be formed of polysilicon. After sequentially forming a second conductive pattern **142**, a third conductive pattern **143**, and first to third capping patterns **146**, **147**, and **148**, respectively, on a first conductive pattern **141**, the first to third conductive patterns **141**, **142**, and **143** may be sequentially etched using the first to third capping patterns **146**, **147**, and **148** as an etching mask. As a result, a bit line structure BLS, including a bit line BL including the first to third conductive patterns **141**, **142**, and **143** and a bit line capping pattern BC including the first to third capping patterns **146**, **147**, and **148**, may be formed.

[0079] A spacer structure SS may be formed on side surfaces (extending in the Z-direction) of the bit line structure BLS. The spacer structure SS may be formed of a plurality of layers. Fence insulating patterns **154** may be formed between the spacer structures SS. The fence insulating patterns **154** may include silicon nitride or silicon oxynitride, although embodiments are not limited thereto. An opening, exposing the second impurity region **105b**, may be formed by performing an anisotropic etching process using the fence insulating patterns **154** and the third capping pattern **148** as an etching mask.

[0080] A lower conductive pattern **150** may be formed on a lower portion of the opening. The lower conductive pattern **150** may be formed of a semiconductor material such as polysilicon. For example, the lower conductive pattern **150** may be formed by forming a polysilicon layer, filling the opening, and then performing an etch-back process.

[0081] A metal-semiconductor compound layer **155** may be formed on the lower conductive pattern

150. The formation of the metal-semiconductor compound layer **155** may include a metal layer deposition process and a heat treatment process, to thereby form a metal silicide.

[0082] An upper conductive pattern **160** may be formed on an upper portion of the opening.

Forming the upper conductive pattern **160** may include sequentially forming a barrier layer **162** and a conductive layer **164**. Thereafter, a patterning process may be performed on the barrier layer **162** and the conductive layer **164** to form insulating patterns **165** passing therethrough. Accordingly, a lower structure, including the substrate **101**, a word line structure WLS, and the bit line structure BLS, may be formed.

[0083] An etch stop layer **168** may be conformally formed on the lower structure, and mold layers **118** and preliminary support layers SP1', SP2', and SP3' may be alternately stacked on the etch stop layer **168**. The mold layers **118** and the preliminary support layers SP1', SP2', and SP3' may be included in the mold structure ST. The etch stop layer **168** may include the mold layers **118** and an insulating material having an etch selectivity under a specific etch condition, for example, at least one of silicon oxide, silicon nitride, silicon carbide, silicon oxycarbide, and silicon carbonitride. The mold layers **118** may be formed of silicon oxide, and the preliminary support layers SP1', SP2', and SP3' may be formed of at least one of silicon nitride, silicon oxynitride, silicon carbonitride, and silicon boron nitride.

[0084] Referring to FIGS. **9** and **11**, a plurality of holes H, passing through the mold structure ST, may be formed (**S20**). An anisotropic etching process may be performed to form the plurality of holes H, and the etch stop layer **168** may serve as a stopper for stopping an etching process. The plurality of holes H may pass through the etch stop layer **168** to expose upper conductive patterns **160**. The plurality of holes H, regions in which lower electrodes **170** are to be formed, may be spaced apart from each other at predetermined intervals on a plane so as to be formed in a regular arrangement, as illustrated in FIG. **1**.

[0085] Referring to FIGS. **9** and **12**, lower electrodes **170** may be formed in the plurality of holes H (see FIG. **11**) (**S30**). The lower electrodes **170** may be formed by depositing a conductive material to fill the plurality of holes H and etching back or planarizing the conductive material to expose an upper surface of the mold structure ST.

[0086] Referring to FIGS. **9** and **13**, support layers SP1, SP2, and SP3, including a support hole SH, may be formed by removing a portion of the mold structure ST (**S40**). In an example embodiment, the support layers SP1, SP2, and SP3 may include support holes SH having the same pattern, but the present inventive concept is not limited thereto. A side surface of the lower electrode **170** may be exposed by selectively removing the mold layers **118** (see FIG. **12**) between the support layers SP1, SP2, and SP3. In example embodiments, a process of removing the mold layers **118** may be performed, for example, using a wet etching process using an etchant (for example, a hydrogen fluoride (HF) solution).

[0087] Referring to FIGS. **9**, **14**, and **15**, a passivation layer PL and an interface layer **175** may be formed (**S50** and **S60**). The passivation layer PL may be formed on the support layer SP. In an example embodiment, a lower passivation layer PLL may be formed on the etch stop layer **168**. Forming the passivation layer PL may include supplying a halogen element, and the passivation layer PL may include a halogen compound. In an example embodiment, the passivation layer PL may be selectively deposited only on a surface of the support layer SP, but the present inventive concept is not limited thereto. In some example embodiments, as illustrated in FIG. **7**, the passivation layer PL may also be formed on surfaces of the lower electrodes **170**.

[0088] The interface layer **175** may be formed on the lower electrodes **170**. In an example embodiment, the interface layer **175** may be selectively deposited only on the surfaces of the lower electrodes **170**, but the present inventive concept is not limited thereto. In some example embodiments, as illustrated in FIG. **8**, the interface layer **175** may also be formed on a surface of the passivation layer PL.

[0089] Referring to FIGS. **9**, **16**, and **2**, a dielectric layer **180** and an upper electrode **190** may be

sequentially formed on the passivation layer PL and the interface layer **175** (**S60**). The dielectric layer **180** may cover an upper surface and a lower surface of each of the passivation layer PL and the interface layer **175**.

[0090] A semiconductor device **100** may be manufactured by forming the upper electrode **190** on the dielectric layer **180**. The upper electrode **190** may fill a space between the lower electrodes **170**.

[0091] FIG. **17** is a flowchart of a method of forming an interface layer according to an example embodiment.

[0092] Referring to FIG. **17**, forming a passivation layer PL and an interface layer **175** (**S50**) may include forming the passivation layer PL by providing a halogen-containing material on a support layer SP (**S52**), forming an interface material layer on the lower electrodes **170** (**S54**), when the passivation layer PL and the interface material layer have a desired thickness, annealing the interface material layer (**S56**), and partially etching the interface material layer (**S58**). When the passivation layer PL and the interface material layer do not have a desired thickness, the process of forming the passivation layer PL (**S52**) and forming the interface material layer (**S54**) are repeated until the desired thickness is reached.

[0093] FIGS. **18** to **20** are schematic vertical cross-sectional views of sequential intermediate processes of a method of forming an interface layer according to an example embodiment.

[0094] Referring to FIGS. **17** and **18**, a passivation layer PL may be formed by providing a halogen-containing material on a support layer SP (**S52**). For example, the halogen-containing material may include at least one of fluorine (F), chlorine (Cl), bromine (Br), and iodine (I). The halogen-containing material may be provided on the support layer SP in a liquid or gaseous state. For example, the halogen-containing material may include at least one of NF₃, HF, and ClF₃. When the halogen-containing material is a gas, the halogen-containing material may be provided in a thermal or plasma atmosphere, for example, of about 215° C. to about 460° C. The passivation layer PL may include a halogen compound.

[0095] Referring to FIGS. **17** and **19**, an interface material layer **175p1** may be formed on lower electrodes **170** (**S54**). In an example embodiment, an interface material layer **175p2** may also be formed on the passivation layer PL. However, since the passivation layer PL includes the halogen compound, a deposition rate of the interface material layer **175p2** on the passivation layer PL may be less than a deposition rate of the interface material layer **175p1** on the lower electrodes **170**. For example, a cross-sectional thickness of the interface material layer **175p2** may be less than a cross-sectional thickness of the interface material layer **175p1**. In some example embodiments, the interface material layer **175p2** may not be formed on the passivation layer PL, and the interface material layer **175p1** may be selectively formed only on the lower electrodes **170**.

[0096] Subsequently, the interface material layers **175p1** and **175p2** may be annealed (**S56**). Elements, included in the interface material layer **175p1**, may permeate into the lower electrodes **170** using an annealing process. For example, as illustrated in FIG. **4**, a second layer **172** of the lower electrode **170** may be formed using the annealing process.

[0097] Referring to FIGS. **17** and **20**, the interface material layers **175p1** and **175p2** (see FIG. **19**) may be partially etched (**S58**). The interface material layer **175p1** (FIG. **19**) may be partially etched on the lower electrodes **170** to form the interface layer **175** using an etching process. The interface material layer **175p2** (FIG. **19**) on the passivation layer PL may be removed using the etching process.

[0098] As illustrated in FIGS. **17** to **20**, the passivation layer PL may be selectively formed on the support layer SP, such that a cross-sectional thickness of the interface layer **175** formed on the support layer SP may need to be reduced, or the interface layer **175** on the support layer SP may not be formed, thereby reducing or preventing electrical bridging in which the lower electrodes **170** are electrically connected to each other by the interface layer **175**.

[0099] As illustrated in FIG. **17**, forming the passivation layer PL (**S52**) and forming the interface material layers **175p1** and **175p2** (**S54**) may be performed a plurality of times until the passivation

layer PL or the interface material layers **175p1** and **175p2** has a desired thickness. In some example embodiments, the interface material layer **175p2** may not be formed on the passivation layer PL, and the process (**S58**) of partially etching the interface material layer **175p1** may be omitted.

[0100] FIG. **21** is a flowchart of a method of forming an interface layer (**S50**) according to an example embodiment.

[0101] Referring to FIG. **21**, unlike the method of forming an interface layer illustrated in FIG. **17**, providing a halogen-containing material on a support layer SP to form a passivation layer PL (**S52**), forming an interface material layer on lower electrodes **170** (**S54**), annealing the interface material layer (**S56**), and partially etching the interface material layer (**S58**) may be performed a plurality of times until the passivation layer PL or an interface layer **175** has a desired thickness.

[0102] FIG. **22** illustrates concentrations of an interface layer according to comparative examples and examples of the present inventive concept. “C1” may refer to a concentration of an interface layer **175** on a surface of a support layer SP, measured, for example, using an X-ray photoelectron spectroscopy (XPS). “C2” may refer to a concentration of an interface layer **175** on a surface of a lower electrode **170** measured, for example, using the XPS.

[0103] Referring to FIG. **22**, in Comparative Example 1, a process of depositing an interface layer **175** without forming a passivation layer PL was performed twice. In Examples 1, 2, and 3 of the present inventive concept, a process of forming a passivation layer PL and depositing an interface layer **175** was performed twice. A thickness of the passivation layer PL in Example 2 may be greater than a thickness of the passivation layer PL in Example 1, and may be less than a thickness of the passivation layer PL in Example 3.

[0104] A concentration of an interface layer **175** on a surface of an support layer SP in Examples 1 to 3 may be low, as compared to Comparative Example 1, thereby reducing or preventing electrical bridging between lower electrodes **170**. In addition, as a thickness of the passivation layer PL increases, the concentration of the interface layer **175** on the surface of the support layer SP may decrease.

[0105] In Comparative Example 2, a process of depositing an interface layer **175** without forming a passivation layer PL was performed three times. In Examples 4, 5, and 6 of the present inventive concept, a process of forming a passivation layer PL and depositing an interface layer **175** was performed three times. A thickness of the passivation layer PL in Example 5 may be greater than a thickness of the passivation layer PL in Example 4, and may be less than the thickness of the passivation layer PL in Example 6.

[0106] A concentration of an interface layer **175** on a surface of an support layer SP in Examples 4 to 6 may be low, as compared to Comparative Example 2, thereby reducing or preventing electrical bridging between lower electrodes **170**. In addition, as a thickness of the passivation layer PL increases, a concentration of the interface layer **175** on the surface of the support layer SP may decrease.

[0107] FIG. **23** illustrates defect rates according to comparative examples and examples of the present inventive concept. “Capacitance” may refer to a capacitance of an information storage structure CAP, and “N” may refer to a defect rate of the information storage structure CAP, for example, the number of times electrical bridging occurs between lower electrodes **170**.

[0108] Referring to FIG. **23**, as compared to Comparative Example 1, electrical bridging between lower electrodes **170** in Examples 1 to 3 may be reduced. In addition, as a thickness of a passivation layer PL increases, a defect rate of an information storage structure CAP may decrease.

[0109] As compared to Comparative Example 2, electrical bridging between lower electrodes **170** in Examples 4 to 6 may be reduced. In addition, as a thickness of a passivation layer PL increases, a defect rate of an information storage structure CAP may decrease and a capacitance of the information storage structure CAP may increase.

[0110] FIG. **24** is a schematic plan view of an integrated circuit device according to example embodiments. FIG. **25** is a vertical cross-sectional view taken along lines X1-X1' and Y1-Y1' of

the integrated circuit device illustrated in FIG. 24.

[0111] Referring to FIGS. 24 and 25, an integrated circuit device 200 may include a substrate 210, a plurality of first conductive lines 220, a channel layer 230, a gate electrode 240, a gate insulating layer 250, and an information storage structure 280. The integrated circuit device 200 may be a memory device including a vertical channel transistor (VCT). The vertical channel transistor may refer to a structure in which a channel length of the channel layer 230 extends from the substrate 210 in a vertical direction (i.e., Z-direction).

[0112] A lower insulating layer 212 may be disposed on the substrate 210. On the lower insulating layer 212, the plurality of first conductive lines 220 may be spaced apart from each other in a first direction (X-direction) and may extend in a second direction (Y-direction). A plurality of first insulating patterns 222 may be disposed on the lower insulating layer 212 to fill a space between the plurality of first conductive lines 220. The plurality of first insulating patterns 222 may extend in the second direction (Y-direction), and upper surfaces of the plurality of first insulating patterns 222 may be disposed at a level the same as that of each of upper surfaces of the plurality of first conductive lines 220; that is, the upper surfaces of the first insulating patterns 222 may be coplanar with the upper surfaces of the first conductive lines 220, relative to an upper surface of the substrate 210 as a reference layer. The plurality of first conductive lines 220 may function as bit lines of the integrated circuit device 200.

[0113] In example embodiments, the plurality of first conductive lines 220 may include doped polysilicon, a metal, a conductive metal nitride, a conductive metal silicide, a conductive metal oxide, or combinations thereof, although embodiments are not limited thereto. For example, the plurality of first conductive lines 220 may include doped polysilicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NbN, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, IrO.sub.x, RuO.sub.x, or combinations thereof, but the present inventive concept is not limited thereto. The plurality of first conductive lines 220 may include a single layer or multiple layers formed of the above-described materials. In example embodiments, the plurality of first conductive lines 220 may include a two-dimensional (2D) semiconductor material. For example, the 2D semiconductor material may include graphene, carbon nanotubes, or a combination thereof.

[0114] The channel layer 230 may be arranged on the plurality of first conductive lines 220 in a matrix form of being spaced apart from each other in the first direction (X-direction) and the second direction (Y-direction). The channel layer 230 may have a first width in the first direction (X-direction) and a first height in a third direction (Z-direction), and the first height may be greater than the first width. For example, the first height may be about 2 to 10 times the first width, but the present inventive concept is not limited thereto. A bottom portion of the channel layer 230 may function as a first source/drain region (not illustrated), an upper portion of the channel layer 230 may function as a second source/drain region (not illustrated), and a portion of the channel layer 230 between the first and second source/drain regions may function as a channel region (not illustrated). The first source/drain region and the second source/drain region may be vertically spaced apart from each other, and the channel region may be a vertical channel region.

[0115] In example embodiments, the channel layer 230 may include an oxide semiconductor. For example, the oxide semiconductor may include In.sub.xGa.sub.yZn.sub.zO, In.sub.xGa.sub.ySi.sub.zO, In.sub.xSn.sub.yZn.sub.zO, In.sub.xZn.sub.yO, Zn.sub.xO, Zn.sub.xSn.sub.yO, Zn.sub.xO.sub.yN, Zr.sub.xZn.sub.ySn.sub.zO, Sn.sub.xO, Hf.sub.xIn.sub.yZn.sub.zO, Ga.sub.xZn.sub.ySn.sub.zO, Al.sub.xZn.sub.ySn.sub.zO, Yb.sub.xGa.sub.yZn.sub.zO, In.sub.xGa.sub.yO, or combinations thereof. The channel layer 230 may include a single layer or multiple layers of the oxide semiconductor. In some examples, the channel layer 230 may have a bandgap energy greater than that of silicon. For example, the channel layer 230 may have a bandgap energy of about 1.5 eV to about 5.6 eV. For example, the channel layer 230 may have optimal channel performance when the channel layer 230 has a bandgap energy of about 2.0 eV to 4.0 eV. For example, the channel layer 230 may be polycrystalline or

amorphous, but the present inventive concept is not limited thereto. In example embodiments, the channel layer **230** may include a 2D semiconductor material. For example, the 2D semiconductor material may include graphene, carbon nanotubes, or a combination thereof.

[0116] The gate electrode **240** may extend on opposite sidewalls of the channel layer **230** in the first direction (X-direction). The gate electrode **240** may include a first sub-gate electrode **240P1** opposing a first sidewall of the channel layer **230**, and a second sub-gate electrode **240P2** opposing a second sidewall opposite to the first sidewall of the channel layer **230**. As one channel layer **230** is disposed between the first sub-gate electrode **240P1** and the second sub-gate electrode **240P2**, the integrated circuit device **200** may have a dual-gate transistor structure. However, the present inventive concept is not limited thereto, and the second sub-gate electrode **240P2** may be omitted, and only the first sub-gate electrode **240P1** opposing the first sidewall of the channel layer **230** may be formed to implement a single gate transistor structure.

[0117] The gate electrode **240** may include doped polysilicon, a metal, a conductive metal nitride, a conductive metal silicide, a conductive metal oxide, or combinations thereof. For example, the gate electrode **240** may be formed of doped polysilicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NbN, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, IrO.sub.x, RuO.sub.x or combinations thereof, but the present inventive concept is not limited thereto.

[0118] The gate insulating layer **250** may surround a sidewall of the channel layer **230** and may be interposed between the channel layer **230** and the gate electrode **240**. For example, as illustrated in FIG. **14**, the entire sidewall of the channel layer **230** may be surrounded by the gate insulating layer **250**, and a portion of a sidewall of the gate electrode **240** may be in contact with the gate insulating layer **250**. In other example embodiments, the gate insulating layer **250** may extend in a direction of extension of the gate electrode **240** (that is, the first direction (X-direction)), and only two sidewalls opposing the gate electrode **240**, among sidewalls of the channel layer **230**, may be in contact with the gate insulating layer **250**. The term “surround” (or “surrounded,” or like terms), as may be used herein, is intended to broadly refer to an element, structure or layer that extends around, envelops, encircles, or encloses another element, structure or layer on all sides, although breaks or gaps may also be present. Thus, for example, a material layer having voids or gaps therein may still “surround” another layer which it encircles.

[0119] In example embodiments, the gate insulating layer **250** may be formed of a silicon oxide film, a silicon oxynitride film, a high-k dielectric film having a dielectric constant higher than that of the silicon oxide film, or combinations thereof. The high-K dielectric film may be formed of a metal oxide or a metal oxynitride. For example, the high-K dielectric film usable as the gate insulating layer **250** may be formed of HfO.sub.2, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, ZrO.sub.2, Al.sub.2O.sub.3, or combinations thereof, but the present inventive concept is not limited thereto.

[0120] A plurality of second insulating patterns **232** may extend on the plurality of first insulating patterns **222** in the second direction (Y-direction), and the channel layer **230** may be disposed between two adjacent second insulating patterns **232**, among the plurality of second insulating patterns **232**. In addition, between the two adjacent second insulating patterns **232**, a first buried layer **234** and a second buried layer **236** may be disposed in a space between two adjacent channel layers **230**. The first buried layer **234** may be disposed on a bottom portion of the space between the two adjacent channel layers **230**, and the second buried layer **236** may be disposed on the first buried layer **234** to fill a remainder of the space between the two adjacent channel layers **230**. An upper surface of the second buried layer **236** may be disposed at a level the same as that of an upper surface of the channel layer **230**, and the second buried layer **236** may cover an upper surface of the gate electrode **240**. Alternatively, the plurality of second insulating patterns **232** may be formed as a material layer continuous with the plurality of first insulating patterns **222**, or the second buried layer **236** may be formed as a material layer continuous with the first buried layer **234**.

[0121] A storage contact **260** may be disposed on the channel layer **230**. The storage contact **260** may vertically overlap the channel layer **230**, and may be arranged in a matrix form of being spaced apart from each other in the first direction (X-direction) and the second direction (Y-direction). The storage contact **260** may be formed of Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NbN, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, IrO.sub.x, RuO.sub.x, or combinations thereof, but the present inventive concept is not limited thereto. The upper insulating layer **262** may surround a sidewall of the storage contact **260** on the plurality of second insulating patterns **232** and the second buried layer **236**.

[0122] An etch stop film **270** may be disposed on the upper insulating layer **262**, and the information storage structure **280** may be disposed on the etch stop film **270**. The information storage structure **280** may include a lower electrode **282**, a dielectric layer **284**, and an upper electrode **286**.

[0123] The lower electrode **282** may pass through (i.e., extend in) the etch stop film **270** to be electrically connected to an upper surface of the storage contact **260**. The lower electrode **282** may be formed as a pillar-type electrode extending in the third direction (Z-direction), but the present inventive concept is not limited thereto. In example embodiments, the lower electrode **282** may vertically overlap the storage contact **260**, and may be arranged in a matrix form of being spaced apart from each other in the first direction (X-direction) and the second direction (Y-direction). Alternatively, a landing pad (not explicitly illustrated, but implied) may be further disposed between the storage contact **260** and the lower electrode **282**, such that the lower electrode **282** may be arranged to have a hexagonal shape.

[0124] In the integrated circuit device **200**, the information storage structure **280** may include a support layer SP (corresponding to SP in FIG. 2), a passivation layer PL (corresponding to PL in FIG. 2), a lower passivation layer PLL (corresponding to PLL in FIG. 2), and an interface layer **175** (corresponding to **175** in FIG. 2). The storage contact **260** may be referred to as a conductive region, and may be electrically connected to the lower electrode **282** of the information storage structure **280**.

[0125] According to example embodiments of the present inventive concept, a passivation layer may be disposed on a support layer, thereby suppressing formation of an interface layer on the support layer. Accordingly, electrical bridging of lower electrodes may be reduced or prevented.

[0126] While example embodiments have been shown and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present inventive concept as defined by the appended claims.

Claims

1. A semiconductor device, comprising: a lower structure having a conductive region; and an information storage structure on the lower structure, wherein the information storage structure includes: a lower electrode electrically connected to the conductive region; a support layer on a side surface of the lower electrode; a dielectric layer on the lower electrode and the support layer; an interface layer between the lower electrode and the dielectric layer; passivation layers between the support layer and the dielectric layer; and an upper electrode on the dielectric layer.
2. The semiconductor device of claim 1, wherein the passivation layers include a material different from that of the support layer.
3. The semiconductor device of claim 2, wherein the passivation layers include a halogen compound.
4. The semiconductor device of claim 1, wherein the interface layer includes niobium oxide and/or tantalum oxide.
5. The semiconductor device of claim 1, wherein at least one of the passivation layers is in contact with the side surface of the lower electrode.

6. The semiconductor device of claim 1, wherein the interface layer has a first portion on an upper surface of the lower electrode, the first portion extending in a horizontal direction parallel to an upper surface of the lower structure, and a second portion in contact with the support layer, the second portion extending in a vertical direction perpendicular to the upper surface of the lower structure.
7. The semiconductor device of claim 6, wherein a side surface of the first portion of the interface layer is in contact with at least one of the passivation layers.
8. The semiconductor device of claim 6, wherein an upper surface and/or a lower surface of the second portion of the interface layer is in contact with at least one of the passivation layers.
9. The semiconductor device of claim 1, further comprising: an etch stop layer in contact with a lower region of the side surface of the lower electrode; and a lower passivation layer between the etch stop layer and the dielectric layer.
10. The semiconductor device of claim 9, wherein an upper surface of the lower passivation layer is in contact with the interface layer.
11. The semiconductor device of claim 1, wherein a cross-sectional thickness of the passivation layers decreases as a distance from the lower electrode decreases.
12. The semiconductor device of claim 11, wherein a portion of the interface layer at least partially overlaps the passivation layer in a vertical direction perpendicular to an upper surface of the lower structure.
13. The semiconductor device of claim 11, wherein a cross-sectional thickness of the interface layer decreases as a distance from the support layer decreases.
14. The semiconductor device of claim 1, wherein a portion of the passivation layers is between the lower electrode and the interface layer.
15. The semiconductor device of claim 1, wherein a portion of the interface layer is between the passivation layers and the dielectric layer.
16. A semiconductor device, comprising: a lower structure having conductive regions; and an information storage structure on the lower structure, wherein the information storage structure includes: lower electrodes electrically connected to the conductive regions; a support layer between the lower electrodes, the support layer having a support opening; a dielectric layer on the lower electrodes and the support layer; interface layers between the lower electrodes and the dielectric layer; a passivation layer between the support layer and the dielectric layer; and an upper electrode on the dielectric layer, wherein in a plan view, the passivation layer extends along a side surface of the support layer exposed by the support opening, between the lower electrodes, and the interface layers extend along side surfaces of the lower electrodes exposed by the support opening.
17. The semiconductor device of claim 16, wherein, in plan view, the passivation layer has an arc shape.
18. The semiconductor device of claim 16, wherein each of the side surfaces of the lower electrodes has a first portion in contact with the support layer, a second portion in contact with the passivation layer, and a third portion in contact with one of the interface layers.
19. The semiconductor device of claim 16, wherein, in a plan view, the interface layers are on a portion of the passivation layer and the passivation layer is spaced apart from the dielectric layer.
20. A semiconductor device, comprising: a lower structure having a conductive region; and an information storage structure on the lower structure, wherein the information storage structure includes: a lower electrode electrically connected to the conductive region; a first support layer and a second support layer on a side surface of the lower electrode, the second support layer below the first support layer; a dielectric layer on the lower electrode, the first support layer, and the second support layer; an interface layer between the lower electrode and the dielectric layer; a first passivation layer between the first support layer and the dielectric layer; a second passivation layer between the second support layer and the dielectric layer; and an upper electrode on the dielectric layer, and wherein the interface layer has a first portion in contact with an upper surface of the

lower electrode and the first passivation layer, and a second portion between the first support layer and the second support layer.
